NLU - Lab 5

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1. Introduction

This work addresses joint intent classification and slot filling on the ATIS dataset. In Part A, I incrementally improved a GPT-2-style decoder trained from scratch by tuning its learning rate and architectural hyperparameters. I also evaluated dropout before the final prediction layers. In Part B, I adapted the joint model to fine-tune pre-trained GPT-2 and BERT checkpoints. The main additional challenge was aligning word-level slot annotations with the sub-tokens produced by the pre-trained tokenizers.

2. Implementation details

Part A. The model predicts one slot label for each token and one intent label for the complete utterance. Since the GPT-2-style backbone is a causal decoder and does not provide a dedicated classification token, I appended a special `CLS` token at the end of each sequence and used its final hidden representation for intent classification. Placing the token at the end ensures that its representation has processed the complete utterance. The training objective is the sum of the slot-filling and intent-classification cross-entropy losses. Starting from a fixed baseline architecture, I first searched for a suitable learning rate. I then tuned `d_model`, `n_heads`, `num_layers`, and `ff_dim` greedily, changing one parameter at a time and carrying the best configuration forward to the next step. Since the resulting model was larger than the initial baseline, I repeated the learning-rate search. I also evaluated the effect of applying dropout to the final hidden representations before passing them to the slot-filling and intent-classification output layers.

Part B. I fine-tuned BERT [2] and GPT-2 [3] for joint intent classification and slot filling. For both models, the training objective is the sum of the intent-classification and slot-filling cross-entropy losses. However, the two tasks must be handled differently because BERT is a bidirectional encoder, whereas GPT-2 is a causal decoder.

Intent classification For BERT, I use the final hidden representation of the special `[CLS]` token. Since BERT uses bidirectional attention, this representation can incorporate information from the complete utterance. GPT-2 does not provide an equivalent classification token. I therefore append an `[EOS]` token to each sequence and use its final hidden representation, which has processed all preceding tokens and thus the complete utterance.

Sub-tokenization The ATIS annotations are defined at the word level, while the pretrained tokenizers may divide a word into multiple sub-tokens. To preserve the word-level labeling scheme, I compute the slot-filling loss using only one sub-token for each original word. The remaining sub-tokens, together with padding and special-token positions, receive the ignore index -100 and do not contribute to the loss. Following [4], I use the **first sub-token** for BERT, since its bidirectional representation can already incorporate the complete word and its surrounding context. For GPT-2, I instead use the **last sub-token**: due to causal attention, this is the first position whose representation has processed every piece of the word. The process is illustrated in the following figure:

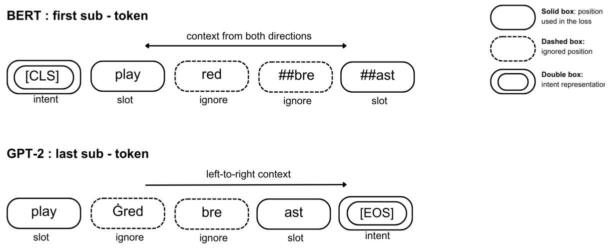

Figure 1: ***Part B*** *- schematic comparison of the positions used for intent classification and slot filling. BERT uses* `[CLS]` *and the first sub-token of each word, whereas GPT-2 uses the appended* `[EOS]` *token and the last sub-token of each word.*

Optimization and robustness Joint training minimizes the sum of the intent-classification and slot-filling cross-entropy losses. The slot loss is computed only at the selected sub-token positions; padding, special tokens, and unused sub-tokens are ignored using index -100. Attention masks prevent padding from affecting the contextual representations. Each experiment was repeated across five random seeds with a fixed train-development split. Early stopping was based on development slot F1, and test results are reported as mean $\pm$ standard deviation across the five runs.

3. Results

The reported values are expressed as mean $\pm$ standard deviation across five runs with different random seeds.

Part A. Table 1 reports the incremental experiments. After hyper-parameter fine tuning, applying dropout with $p = 0.3$ produced the best results.

Table 1: ***Part A*** *- incremental experiments. The configuration column reports* `d_model` / `n_heads` / `num_layers` / `ff_dim`.

Experiment	LR	Configuration	Dev Slot F1	Dev Intent Acc.
Baseline	$5 \cdot 10^{-3}$	20 / 1 / 1 / 20	0.9338 ± 0.0054	0.9550 ± 0.0190
Greedy tuning	$5 \cdot 10^{-3}$	80 / 1 / 3 / 80	0.9468 ± 0.0039	0.9715 ± 0.0041
LR readjustment	$5 \cdot 10^{-3}$	80 / 1 / 3 / 80	0.9468 ± 0.0039	0.9715 ± 0.0041
Dropout ($p = 0.3$)	$5 \cdot 10^{-3}$	**80 / 1 / 3 / 80**	$\mathbf{0.9556 \pm 0.0034}$	$\mathbf{0.9767 \pm 0.0030}$

Part B. Table 2 compares the two pre-trained backbones. BERT achieved the best results, improving the slot-filling F1 score by approximately 2.71 percentage points over GPT-2. The improvement is consistent with the bidirectional context available to BERT, which is particularly useful when predicting a slot label for each word.

Table 2: ***Part B*** *- comparison of pre-trained models on ATIS.*

Model	Test Slot F1	Test Intent Acc.
GPT-2 (base)	0.9298 ± 0.0019	0.9711 ± 0.0069
BERT (base)	$\mathbf{0.9569 \pm 0.0012}$	$\mathbf{0.9756 \pm 0.0013}$

4. References

[1] M. Rizzoli, “Natural language understanding labs,” https://github.com/massimo-rizzoli/NLU-2026-Labs, 2026, university of Trento, accessed: 2026-06-02.

[2] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*. Minneapolis, Minnesota: Association for Computational Linguistics, 2019, pp. 4171–4186.

[3] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, “Language models are unsupervised multitask learners,” OpenAI, Tech. Rep., 2019.

[4] Q. Chen, Z. Zhuo, and W. Wang, “Bert for joint intent classification and slot filling,” *arXiv preprint arXiv:1902.10909*, 2019.